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ORIGINAL RESEARCH

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Brief cycles of lower-limb occlusion accelerate recovery kinetics in soccer players

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ABSTRACT

Objective: The aim of this study was to assess the effect of intermittent vascular occlusion (IVO) on recovery following simulated soccer physical demand test in soccer players.

Methods: Twelve soccer players completed the Loughborough Intermittent Shuttle Test (LIST) in two conditions placebo (PLA) and IVO followed by intermittent lower-limb occlusion. Physical performance (Squat jump: SJ, countermovement jump: CMJ, maximal voluntary contraction: MVC, and 20 m sprint: SP), muscle damage parameters (creatine kinase: CK, Lactate dehydrogenase: LDH), inflammatory parameter (C-reactive protein: CRP), and perceived muscle soreness (DOMS) were assessed before, immediately after (0 h), and 24 h, 48 h, and 72 h following the exercise.

Results: Following the LIST, a decrease was observed in all Physical performance within 48 h in PLA condition ($p < 0.05$), compared to PLA treatment, IVO treatment attenuated the decrease of SJ and CMJ at 24 h and at 48 h and for MVC and SP within 48 h after the LIST ($p < 0.05$). CK and LDH levels increased within 24 h post-exercise in both conditions ($p < 0.05$), but with a lower level in IVO compared to PLA condition ($p < 0.05$). Likewise, DOMS values were significantly lower with IVO condition compared to PLA condition immediately and at 24 h after exercise.

Conclusion: The results of the present study suggest that the application of IVO after simulated soccer physical demand test accelerated recovery kinetics in soccer players.

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KEYWORDS
Ischemia; soccer; fatigue; muscle damage; vascular occlusion

Introduction

Soccer is an intermittent sprint sport that places significant physical demand on players [1,2]. The eccentric nature of the actions performed during a soccer game has the potential to induce muscle damage and leading to impair of physical performance which may last for up to 96 h post games [3]. For some players, the number of games per season is very high with up to 60–80 per season. In this context, the number of weeks with at least two games per week, with a 48 h-96 h-recovery period, is frequent. The insufficient recovery period coupled with a congested schedule has been proposed to explain the increase of injury incidence when two or three games are played in a week [4].

The ability of soccer players to fully recover before their next major competition is crucial not only for their performance but for injury prevention as well. To improve recovery from a previous competition, soccer players might benefit from interventions including stretching, massage, compression, cold-water immersion [5].

Recently, the use of intermittent vascular occlusion (IVO) has come to attention for sport scientists [6,7]. This intervention involves short periods of intermittent vascular occlusion of the limb (2–5 min), followed by reperfusion. IVO consisting of tissue exposure to brief periods of ischemia leads to resistance against cell injury caused by subsequent prolonged ischemia [8]. In addition, it has been recently shown that IVO improves exercise performance [9]. In fact, recent works reported that IVO applied to the lower limbs increase maximal workload during an incremental cycling test [10] and decreased the time to complete a 5-km running trial [11]. When applied to the upper limbs, IVO reduced the time in100and200-m swimming trials [12]. Although exact mechanisms by which IVO improves physical performance are unclear, it has been supposed that IVO is purported to increase blood flow, adenosine concentrations (ATP), and oxygen and nutrient delivery via vasculature dilation [13].

While the beneficial effect of IVO on the physical performance enhancement has been clearly established, the potential role of ischemia and reperfusion cycles in the recovery process improvement following exercise has received little research attention. The first study to demonstrate the IVO potential use in the recovery process from athletic efforts was conducted by Beaven et al. [6]. The authors reported that IVO applied after a training session resulted in beneficial effects on repeated sprint ability and jump height performances. Recently, Page et al. [7] reported that IVO could reduce muscle damage and accelerate the recovery process at 24 h, 48 h, and 72 h following exercise-induced muscle damage.

In the same way, Arriel et al. [14] reported that IVO treatment prevents a decrement in performance 24 h after a maximal incremental cycling exercise test in male cyclist.

The proposed beneficial effects of IVO on recovery kinetics have been attributed to post-occlusion increases in blood flow associated with elevated adenosine levels [13] and activation of ATP-sensitive potassium channels [10], which potentially

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result in elevated blood flow and enhanced skeletal muscle contractile function [6].

Conversely, other findings did not support the beneficial effects of IVO on the recovery process. When compared to a passive recovery from a heavy strength training session IVO did not induce a better recovery of perceived muscle soreness and performance following the exercise, in strength-trained male [15]. Similarly, Williams et al. [16] demonstrated that IVO had no effect on physiological or neuromuscular markers 2 h neither 24 h after sprinting exercise in rugby players. These controversies about the effects of IVO on recovery kinetics might be due to differing treatment protocols, cuff pressures, and timings of recovery assessments.

In addition to these equivocal results, to our knowledge, no study has analyzed the effects of IVO on recovery kinetics following a protocol designed to replicate the soccer game physiological demands in soccer players.

Therefore, the main purpose of this study was to investigate the effect of IVO on recovery indices following the loughborough intermittent shuttle test (LIST), a protocol designed to simulate the activities of real soccer match Nicholas et al. [17]. We hypothesized that brief cycles of lower-limb occlusion would reduce muscle soreness and damage and accelerate the recovery of muscle function.

Methodology

Experimental design

This study was a single-blind (the tester was blind to the protocol interventions), placebo (PLA)-randomized crossover with two sessions separated by 7-day washout. The principal investigator was responsible for the randomizing procedures. Randomization of participants in each condition was conducted by using a random-numbers generator (www.randomization.com).

Each experimental session consisted of a prolonged intermittent test followed by post-exercise tests and the intervention or placebo protocols (Figure 1).

Biochemical, physical, and perceptual markers were before, immediately after (0 h), and at 24, 48, and 72 h after the intermittent test. During the period devoted to each condition, no training session was implemented and participants were requested not to use any different recovery treatments (e.g. massage, compression garments).

The participants were also asked to refrain from any exercise and not to consume alcoholic or caffeinated beverages 24 h prior to each test and to maintain their normal dietary habits during the study. All tests were performed in a quiet laboratory under standardized conditions (temperature: 21–23°C; relative humidity: 40–60%) and were performed at the same time of day.

Participants

Twelve male soccer players (age = 23 ± 1 yr., height = 179 ± 1,22 cm, mass = 77,9 ± 3,42 kg) from a semiprofessional Tunisian senior squad volunteered to participate in this study. All players were members of the same team, and they had a mean of 7.2 years' experience playing. They are also regulatory performed between 4 and 5 soccer sessions training per week with one official match. Experimental testing sessions took place in the late off-season to the early pre-season phase of the players' training year. The study was conducted in accordance with the Declaration of Helsinki and was approved by the local clinical Research Ethics Committee (CPP N°: 0104/18). After receiving a description of the protocol, and having been made aware of the possible risks and benefits associated with the study, each subject signed a written informed consent prior to participation.

Materials and methods

Physical performance tests

For each session, the order of physical performance tests was identical throughout the protocol: maximal voluntary contraction of quadriceps (MVC), squat jump (SJ), countermovement jump (CMJ), and 20-m sprint (SP).

Initially, participants performed the same standardized 15-min warm-up comprising easy jogging, 3–5 accelerations, movement flow consisting of bodyweight squat variations, lunges, and hip mobility drills.

Subsequently, participants performed brief isometric maximal voluntary contractions (MVC) of knee extension with 90° flexion on an isometric dynamometer (Good Strength, Metitur, Finland). Before MVC participants performed specific warm-up consisting of three submaximal repetitions of knee extension, starting from 90° flexion. For MVC, players completed one familiarization repetition of MVC following by 5-min recovery, before they completed two maximal repetitions of knee-extension with knee flexed at 90°, between the two trial participants rested for 60-s. During testing, participants were

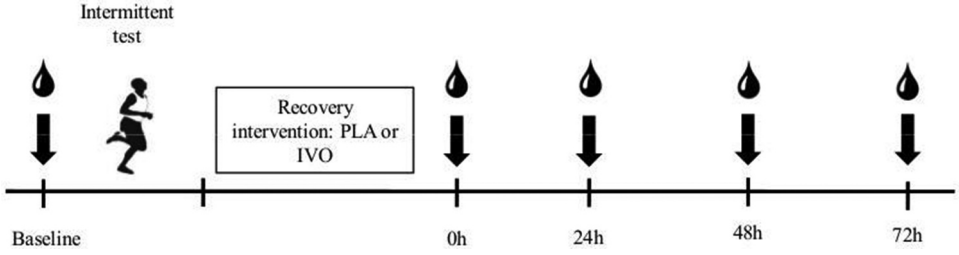

Figure 1. Overview of the entire experimental period. Drop marks denote the time points when blood samples were taken. Downward arrows denote the time points when physical performance tests and perceptual ratings were recorded.

seated on an isometric dynamometer (Good Strength, Metitur, Finland) and participants were stabilized with safety belts strapped across the chest, thighs, and hips, in order to avoid lateral, vertical, or frontal displacements. All measurements were taken from the participant's dominant leg, with a 90° knee flexion angle from full extension.

Following MVC test, players performed one familiarization of either CMJ and SJ following by 3-min recovery. Each athlete performed three SJ and three CMJ using Opto-jump system (Microgate SARL, Italy), the best jump height of SJ and CMJ was recorded for analysis. During SJ and CMJ, players placed hands on the hips.

Finally, participants performed the 20-m sprint test. For determining sprint performance, we used single beam timing lights (Brower Timing System, IRD-T175, Draper, UT, USA) placed at 0 and 20 m. The two timing lights (start and finish) were mounted at 100 cm, in order to match approximately hip height. Players were asked to start each sprint from a line placed 50 cm before the starting line. Participants began running upon a verbal signal and they were instructed to run as fast as possible to complete the 20-m distance. Players performed one familiarization sprint followed by 3-min recovery. Subsequently, players performed two trials interceded by 3 min of recovery, the best trial was recorded for subsequent analysis.

Simulated soccer physical demand test

The LIST was conducted according to the protocol described by Thompson et al. [18]. Participants were asked to run between markers set 20 m apart at varying speeds dictated by an audio signal. The test consisted of six exercise sets approximately 15 min long separated by periods of 3 min, incorporating walking, sprinting, recovery, cruising, and jogging in intensities between 55% and 95% of an individual´s estimated VO2max and according to the test protocol [18]. Heart rate (HR) average was recorded for each set throughout the intermittent exercise (Polar Team2 Pro, Polar, Finland).

Blood analysis

During each session, venous blood samples were taken from an antecubital vein into a plain vacutainer tube. Whole blood collected into tubes, no additive was allowed to clot for 30 min at room temperature, centrifuged at 3000 rpm for 15 min at 4°C, and then processed for plasma and stored at −70°C until analysis. Plasma creatine kinase activity (CK), lactate dehydrogenase (LDH), and serum C-reactive protein (CRP) were determined spectrophotometrically using commercial test kits (A11A01632, Horiba-ABX, Montpellier, France; myoglobin bioMerieux 30,446) according to the instructions of the manufacturers. The intra-assay coefficients of variation for these parameters were 1.3%, 0.2%, and 1.16%, respectively.

Perceptual scales

Following the LIST, participants rated the perceived exertion (RPE) of the session using the Borg scale. The RPE scale allows athletes to give a subjective exertion rating for the physical exercise. It consists of 11-point scale ranging from 0 (very light) to 10 (very hard)

Muscle soreness questionnaire for quadriceps and hamstring muscles was used, within 10 min from the end and at 24 h, 48 h, and 72 h after the LIST on a scale from 0 ('absence of soreness') to 10 ('very intense soreness').

Body mass fluid loss and blood pressure

For all participants, body height was measured using a stadiometer, and body weight was measured using a calibrated electronic scale (Tanita TBF 401, Tanita Corp., Japan). Total body water and body fat percentage were assessed by using bioelectrical impedance analyzer (Tanita TBF 401, Tanita Corp., Japan). In order to determine sweat loss during the LIST, players were weighed wearing dry shorts before and immediately after the exercise using a digital scale (Tanita TBF 401, Tanita Corp., Japan). Finally, blood pressure was measured with a sphygmomanometer (Microlife USA, Inc, Clearwater, Florida) on the participant's upper right arm. Three measurements of blood pressure were recorded at the end of a 5-min spine rest. The average of the three measurements was used for data analysis.

Recovery protocol

IVO consisted of three cycles of 5-min occlusion separated by 5-min reperfusion [7]. Occlusion was induced by a blood pressure cuff (77.0 cm length x 21.5 cm width). Bilaterally vascular occlusion was applied on the proximal portion of the thigh in a spine position. In IVO, a pressure of 50 mm Hg above the systolic blood pressure was used during occlusion and of 0 mm Hg during the reperfusion [14,19]. However, in PLA, a pressure of 20 mmHg was used during the occlusion period. To ensure the presence of occlusion/reperfusion phases, blood flow of the anterior tibial artery was checked continually by an auscultation [14].

Statistical analyses

All data are presented as means ± standard deviation (SD) and were analyzed using Statistica for Windows software (version 6.0, StatSoft, Inc, Tulsa, OK). Normality of every dependent variable was confirmed using Shapiro–wilk test. Values of physical and biochemical parameters were normalized to 100%. For all physical and biochemical and perceived soreness data, two-way anova has been used [Condition (IVO or PLA) × time (Baseline,0 h, 24 h, 48 h, and 72 h post-exercise)]. The effect of time and condition on the dependent variables was analyzed using a two-way analysis of variance with repeated measures. When appropriate, post hoc comparisons were made with the Bonferroni test. The standardized effect size (Cohen d) analysis was used to interpret the magnitude of differences among measurements. In addition, effect size data (ES) were calculated using the following formula:

$$ES = \frac{SS_{effect}}{SS_{toral}}$$

where: SSeffect is the sum of the square of the effect, and the SStotal is the total sums of squares for all effects. In addition, ES was assessed to determine the magnitude of changes using the following criteria: <0.2 = trivial, 0.2–0.6 = small, 0.6–-1.2 = moderate, 1.2–2.0 = large, and >2.0 = very large (Hopkins 2002). Power calculation was performed using G*Power v.3 (Faul et al., 2009). Statistical significance was accepted at $p < 0.05$. For reliability of physical and biochemical tests, the coefficient of variation (CV), intraclass correlation coefficient (ICC), 90% confidence intervals (CI), and typical error (TE) were calculated.

Results

Sweet loss and heart rate

As shown in Table 1, statistical analysis showed no significant differences between PLA and IVO in fluid loss values (2, 16% and 1, 92%, respectively) ($p = 0.64$), and in heart rate values (168 ± 11 bpm and 167 ± 12 bpm, respectively) ($p = 0.72$) during the LIST protocol.

Recovery kinetics for physical performance

The two-way ANOVA showed a significant main effect of time on the SJ performance ($F_{4,40} = 4.95$, $p < 0.01$, ES = 0.33). No significant time x condition interaction was found ($F_{4,40} = 2.45$, $p = 0.06$, ES = 0.19). SJ decreased immediately after the LIST at 0 h ($p = 0.02$) at 24 h and 48 h ($p = 0.01$) and returned to baseline levels 72 h after the test in PLA. No significant change was observed for SJ performance during the recovery period in IVO (Figure 2(a)).

Likewise, statistical analysis showed a significant main effect of time on the CMJ performance ($_{F4,40} = 5.18$, $p < 0.01$, ES = 0.34). No significant time x condition interaction was found ($F_{4,40} = 1.75$, $p = 0.15$, ES = 0.15).CMJ performance decreased immediately after exercise at 0 h, at 24 h and at 48 h ($p < 0.01$) and returned to baseline levels 72 h after the LIST in PLA, whereas in IVO, it did not change throughout the recovery period (Figure 2(b)).

We observed an interaction between condition and time for MVC ($F_{4,44} = 8.29$, $p < 0.01$, ES = 0.45). Post hoc analysis showed that MVC performance decreased immediately after

Table 1. Descriptive characteristics (Mean ± SD) of participants.

	Placebo condition	Lower-limb occlusion condition
Age (years)	23,05 ± 0,66	-
Weight (kg)	77,9 ± 3,42	76,83 ± 3,29
Height (cm)	179 ± 1,22	-
Body mass index (kg/m^2)	23,25 ± 1,7	23,05 ± 0,8
Body Fat (%)	9,79 ± 2,45	9,92 ± 2,98
Total body water (%)	65,21 ± 1,90	65,76 ± 1,44
Fluid loss (%)	2,16 ± 0,16	1,92 ± 0,14
Heart rate(bpm)	168 ± 11	167 ± 12

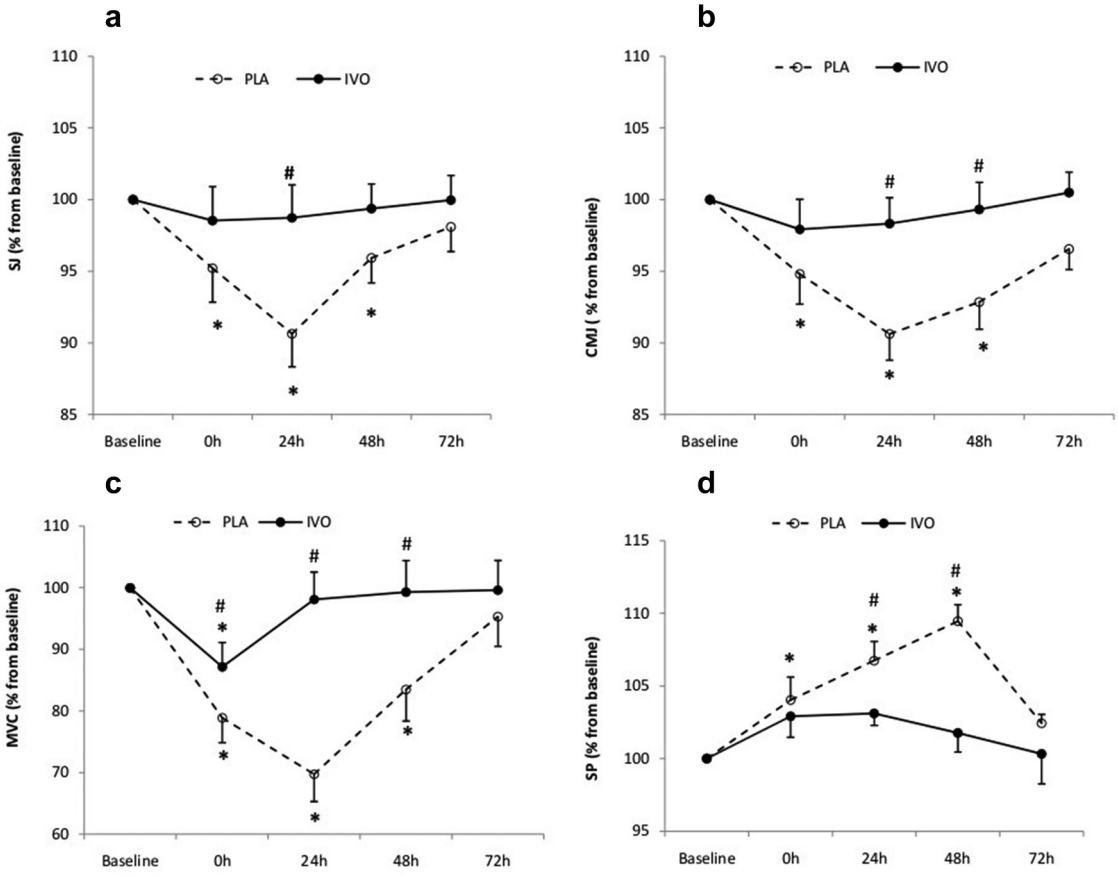

Figure 2. Squat jump (SJ) (a), countermovement jump (CMJ) (b), maximal voluntary contraction of quadriceps (MVC) (c) and 20 meters sprint test (SP) (d) values recorded at baseline and following the intermittent test (at 0 h, 24 h, 48 h and 72 h) during Placebo (PLA) and occlusion session session (OCC). Values are means and standard deviations. *: Significant difference in comparison to baseline ($p < 0.05$). #: Significant difference in comparison to PLA ($p < 0.05$).

exercise at 0 h, 24 h, and at 48 h ($p = 0.01$) and returned to baseline levels 72 h after the test in PLA. However, MVC decreased only at 0 h ($p = 0.01$) with IVO. In addition, MVC values were higher at 0 h, 24 h, and at 48 h after the LIST ($p < 0.05$) in IVO compared to PLA (Figure 2(c)).

For SP, we noted an interaction between condition and time ($F_{4,44} = 2.94$, $p = 0.03$, ES = 0.22). Post hoc analysis showed that SP performance decreased immediately after the LIST 0 h, at 24 h and at 48 h ($p = 0.01$) and returned to baseline levels 72 h after the test in PLA condition. However, no significant change in SP throughout the recovery period in IVO has been observed. Moreover, SP performance was lower at 24 h and at 48 h ($p < 0.05$) in PLA compared to IVO (Figure 2(d)).

Recovery kinetics for biochemical parameters

As shown in Figure 3, statistical analysis showed an interaction between condition and time for CK ($F_{4,44} = 8.58$, $p < 0.01$, ES = 0.46). In both conditions, post hoc analysis showed that CK activity increased at 0 h and at 24 h ($p < 0.01$) after the LIST. However, the increase of CK level was higher at 0 h and at 24 h ($p < 0.01$) in PLA compared to IVO (Figure 3a).

We observed an interaction between condition and time for LDH ($F_{4,44} = 4.00$, $p < 0.01$, ES = 0.28). LDH activity increased at 0 h, 24 h, 48 h and at 72 h ($p < 0.01$) after the LIST in both conditions. However, LDH activity at 0 h was higher in PLA compared to IVO ($p < 0.01$) (Figure 3b).

Statistical analysis showed no interaction between condition and time for CRP activity ($F_{4,44} = 1.97$, $p = 0.11$, ES = 0.16). We noted an effect only of measurement time for CRP ($F_{4,44} = 6.71$, $p < 0.01$, ES = 0.41), which increased only at 24 h ($p < 0.01$) after the intermittent test in both conditions (Figure 3c).

Recovery kinetics for perceptual scales (DOMS)

Statistical analysis showed an interaction between condition and time for DOMS ($F_{4,44} = 5.99$, $p = 0.001$, ES = 0.37). Post hoc analysis showed the DOMS peaked in PLA and IVO at 0 h, at 24 h, and at 48 h ($p < 0.01$). DOMS score was lower at 0 h and at 24 h ($p < 0.01$) in IVO as compared to PLA condition (Figure 3d).

Reliability

The reliability of each outcome measure was evaluated using the results of the beeline test of each condition (Table 2). Intraclass Correlation Coefficient (ICC), typical error, and coefficient of variation are criteria for reliability (Vincent 1999). Reliability is generally considered high when ICC is above 0.90 and good when ICC is above 0.75 (Cohen 1977). A good level of reliability was reported for SJ (ICC = 0.78) CMJ (ICC = 0.76) and MVC (ICC = 0.86) and LDH (ICC = 0.77) and CRP (ICC = 0.75). Reliability for SP (ICC = 0.69) CK (ICC = 0.70) was low. Typical error was low for all the tests, which indicates

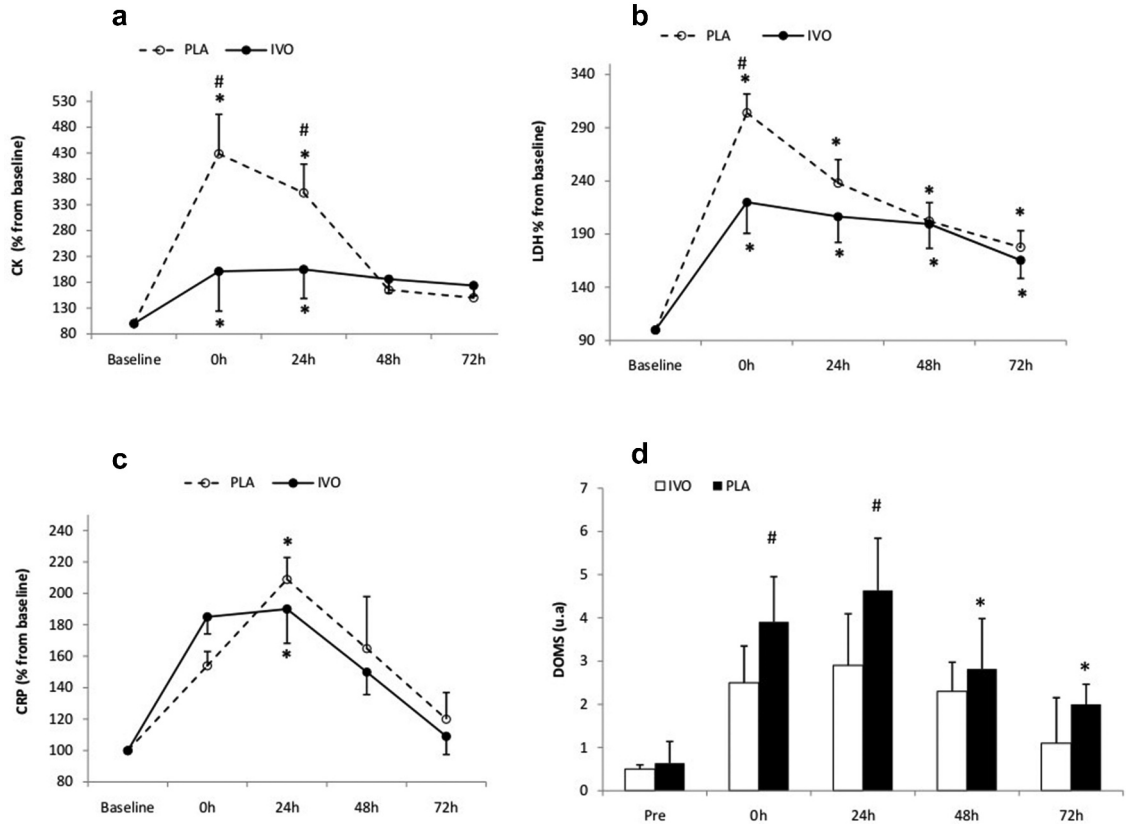

Figure 3. Plasma creatine kinase (CK) (a), Lactate dehydrogenase (LDH) (B), C-reactive protein (CRP) (C) and perceived muscle soreness (D) values recorded at baseline and following the intermittent test (at 0 h, 24 h, 48 h and 72 h) during Placebo (PLA) and occlusion session (OCC). Values are means and standard deviations. *: Significant difference in comparison to baseline ($p < 0.05$). #: Significant difference in comparison to PLA ($p < 0.05$).

Table 2. Reliability of outcomes measured during the experimental protocol.

	Trial 1	Trial 2	Mean of change	Effect Size (90% CI)	TE (90% CI)	ICC (90% CI)	CV
SJ (cm)	30.3 ± 1.4	30.7 ± 1.8	0.4 ± 1.1	0.25 (90% CI = −0.47 to 0.94)	2.89 (90% CI = −2.14 to 4.60)	0.78 (90% CI = 0.45 to 0.92)	CV = 9.2%
CMJ (cm)	31.6 ± 2.2	31.8 ± 3.4	0.20 ± 0.9	0.07 (90% CI = −0.63 to 0.77)	2.65 (90% CI = 1.96 to 4.22)	0.76 (90% CI = −0.55 to 0.85)	CV = 8.2%
MVC (N)	908.4 ± 95.6	924.3 ± 97.9	15 ± 39.2	0.05 (90% CI = −0.65 to 0.75)	2.04 (90% CI = −1.51 to 3.25)	0.84 (90% CI = 0.59 to 0.94)	CV = 7.6%
SP (s)	3.18 ± 0.10	3.16 ± 0.06	0.02 ± 0.13	−0.23 (90% CI = −0,93 to 0.48)	0.10 (90% CI = −0.07 to 0.16)	0.69 (90% CI = 0.30 to 0.88)	CV = 11.2%

Trial 1 and trial 2 are respectively the baseline values of the two conditions. CV: coefficient of variation. ICC: intraclass correlation coefficient. CI: 90% confidence intervals. TE: typical error.

a good reliability as the smaller the typical error the more reliable the measurements. A Coefficient of variation is generally considered high when it is above 10% [20]. For SJ (CV = 9.2%), CMJ (CV = 8.2%) and MVC (CV = 7.6%) coefficients of variation were low. For SP (CV = 11.2%) coefficient of variation was high.

Discussion

The primary aim of the present study was to examine the effect of intermittent lower-limb occlusion on the recovery kinetics following simulated soccer match in soccer players. The findings of the present study confirm our hypothesis that IVO can shorten the recovery processes following simulated soccer match. In fact, IVO promoted a diminished perception of perceived muscle soreness, a lower decrement in physical performance, and a reduced level of muscle damage biomarkers throughout the recovery period.

In the present study, LIST protocol resulted in a similar response of physical, biochemical, and soreness with other studies [17,21]. The eccentric nature of the LIST protocol may contribute to an important amount of fatigue. Fatigue occurs in soccer matches owing to a combination of central and peripheral factors, which leads to a physical performance decrement, especially a decrease in neuromuscular performance and is in accordance with previous studies [21,22].

Results of the present study showed that IVO accelerates post-exercise recovery of MVC and SP performance. These results are in line with those reported by Page et al. [7]. These authors demonstrated that IVO intervention accelerated the recovery of MVC following exercise-induced muscle damage 24 h earlier than the PLA condition. Similarly, Beaven et al. [6] reported faster recovery of CMJ and sprint performances with IVO as compared to control conditions following repeated sprint exercise. Recently, Arriel et al. [14] showed that 2 sessions of 5-min occlusion/5-min reperfusion led to maintain exercise performance 24 h following incremental cycling test in trained cyclists.

The mechanisms explaining faster recovery of physical performance with IVO are not completely elucidated. It has been supposed that the increase in local blood volume after IVO likely contributed to performance improvement during the recovery period [6]. The increased blood volume with IVO possibly accelerated ATP and PCr repletion via increasing O_2 availability and delivery in muscle [23]. In addition, improvement of oxygen delivery is associated with increases in adenosine, activation of ATP-sensitive potassium channels [10], and inhibition of afferent fatigue signaling increasing neural drive [24], which could also possibly account for the improvement in performance during the recovery period.

On the other hand, during soccer match or simulated soccer match play as LIST which including many numbers of accelerations and decelerations, eccentric muscle contractions during these activities are considerable resulting in damaged muscle fiber and therefore cell breakdown. Some markers, such as CK and LDH, have been used to indicate the grade of cell damage [25–27]. Previous studies have investigated the effects of IVO on markers of muscle damage following exercise and the reported results are contradictory [7,16]. This divergence in studies results is mainly due to the differences in the experimental design, pressure cuffs, subject's fitness level, and training background. However, the efficiency of IVO to protect against exercise-induced muscle damage symptoms in soccer players has not been studied before, although indirect evidences imply that IVO intervention may act as a recovery enhancer. Furthermore, it has been shown that intermittent lower-limb occlusion is able to reduce some markers of muscle damage such as CK levels after damaging exercise [7]. Results of the present study demonstrated that IVO intervention could significantly attenuate the increased CK and LDH activity observed during the recovery period. In fact, the mechanisms underlying the ability of IVO to affect the recovery process after one bout of exercise are still speculative. Lower CK and LDH levels may be associated with the effect of IVO on improvements in vascular function, which regulate blood flow to remove these markers of muscle damage for uptake and use [28,29]. In fact, brief cycles of ischemia and reperfusion improve muscle blood flow via increased adenosine levels which induce vascular vasodilatation and increased therefore muscle blood flow. On the other hand, it has been suggested in animal studies that the effect of IVO on adenosine is mediated via interaction A3 adenosine receptors [13]. In fact, stimulation of adenosine A3 receptors has been reported to decrease creatine kinase responses after eccentric exercise protocol [30].

Players in the present study reported reduced perceived muscle soreness during the recovery period compared with PLA treatment. Reductions in muscle soreness and improved perception of recovery may have important implications for physical performance enhancing well-being athlete's state that may translate to greater athletic performance [31,32]. These results are in line with previous clinical research using a similar technique of IVO which reported reduced

postoperative pain following total knee arthroplasty [33]. In the same line, Page et al. [7] reported that IVO reduced perceptions of soreness after exercise-induced muscle damage in active males. The mechanism responsible for the lower perception of soreness with IVO during the recovery period remains unclear. However, it is likely that reduced muscle soreness following LIST was due to the reduction in the inflammatory response associated with exercise. In this context, it has been indicated that the use of 3 × 5 min of occlusion, separated by 5 min of reperfusion immediately before a bout of eccentric exercise, blunts the skeletal muscle damage responses as indicated by a lower magnitude of CK activity and reduced pain and muscle swelling [34]. It should be also noted that brief periods of ischemia following by reperfusion have been shown to increase levels of nitric oxide (NO) [35].In fact, NO plays an important role in reduction of inflammation and muscle repair [36], which could possibly reduce muscle soreness and pain.

Some limitations of the present study warrant mention. First, even we blinded the tester to conditions it was difficult to blind subjects from the conditions due to the obvious difference in cuff pressure. Moreover, another group of control would be useful to identify the impact of occlusion and to determine its effects when compared with another group without cuffs. It should be also taking into account that in the present study we did not measure muscle oxygenation and total serum antioxidants. Such measurements could be the focus of future investigations.

Conclusion

In conclusion, this study is the first to assess the effect of IVO on recovery kinetics following simulated match play in soccer players. Results of the present study clearly showed that the application of 3 cycles of 5 min of occlusion followed by 5 min of reperfusion immediately following simulated soccer match attenuated the increases in muscle soreness and muscle damage markers and accelerates the recovery of neuromuscular functions in semiprofessional soccer players. From a practical view, our findings suggest that IVO offers an attractive method of enhancing recovery.

Practical applications

This study shows that using IVO was more beneficial to reduce muscle soreness and muscle damage markers and accelerates the recovery of neuromuscular functions after a simulated soccer match. Implementing IVO is a potentially useful strategy to accelerate recovery after a simulated soccer match.

Declaration of interest

No potential conflict of interest was reported by the authors.

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